

BioSCape Vegetation Plots, Cape Floristic Region, South Africa, 2023

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Introduction

Dataset title: BioSCape Vegetation Plots, Cape Floristic Region, South Africa, 2023

Summary: Vegetation composition and cover at 189 field plots in the Greater Cape Floristic Region of South Africa, collected for the Biodiversity Survey of the Cape (BioSCape). Each plot was surveyed once between 2023-05-03 and 2023-11-01 and divided into four quadrants (NE, NW, SE, SW). The product reports quadrant-level and plot-level fractional cover of bare soil, bare rock, dead vegetation, and live vegetation, together with species-level percent cover, stem counts, and canopy diameter for 709 species. Line-intercept hits along the north–south and west–east plot diameters are also included. All species with at least 5% cover were included in the dataset, plus additional species where needed to make up 80% of the total cover. The data were collected in the field by botanists. Plot-center coordinates are in the companion spatial file.

Dataset overview

Project: Biodiversity Survey of the Cape (BioSCape), an integrated field and airborne campaign in the Greater Cape Floristic Region (Cardoso et al. 2025).

Related publication:

Cardoso, A.W., E.L. Hestir, J.A. Slingsby, C.J. Forbes, G.R. Moncrieff, W. Turner, A.L. Skowno, J. Nesslage, P.G. Brodrick, K.D. Gaddis, and A.M. Wilson. 2025. The biodiversity survey of the Cape (BioSCape), integrating remote sensing with biodiversity science. *npj Biodiversity* 4:2. <https://doi.org/10.1038/s44185-024-00071-5>

Related datasets:

- Fitzpatrick, M.C., D. Spalink, A.J. Elmore, J. Measey, and S. Kritzing-Klopper. 2026. BioSCape: Vegetation Composition at Plots with Woody Invasives, CFR, South Africa. ORNL DAAC. <https://doi.org/10.3334/ORNLDAAC/2511>. Those plots follow this survey protocol and were placed near these plots, in vegetation that also contained woody invasive cover.
- Slimp, M., J. Malindi, R. Meyer, A. Cunningham, J. Rourke, M. Hayden, A. Hargey, J. Nesslage, S. Johnson, M. Rossi, and N. Stavros. 2025. BioSCape Vegetation Surveys Berg and Eerste River Catchments, South Africa, 2022–2023. ORNL DAAC. <https://doi.org/10.3334/ORNLDAAC/2425>. Same campaign, different sites and some different measurements.

Acknowledgments: Cardoso et al. (2025) report NASA award 80NSSC21K0086 to A.M. Wilson and E.L. Hestir for the U.S. components of BioSCape, and National Research Foundation grant 142438 to J.A. Slingsby.

Dataset characteristics

Spatial coverage: Greater Cape Floristic Region, Western Cape and Eastern Cape, South Africa. Site codes in the plot summary: agulhas, anysberg, bainskloof, bavianskloof, bontebok, capepoint, cederberg, dehoop, franschoekpass, gardenroute, grootbos, grootwinterhoek, hawequas, houwhoekpass, kogelberg, langeberg, outeniqua, rooiberg, swartberg, viljoenspass, vrolijkheid, westcoast.

Spatial resolution: One circular vegetation plot with a diameter of 10m at each location, subdivided into four quadrants.

Temporal coverage: 2023-05-03 00:00:00 to 2023-11-01 23:59:59 UTC. **date** in the quadrat summary is the survey calendar date. Each plot has one date.

Temporal resolution: One sample per plot.

Study area: Bounding box of the `plot_centers` layer in `bioscape_veg_spatial_v20261007.gpkg`, WGS 84 decimal degrees.

Site	Westernmost longitude	Easternmost longitude	Northernmost latitude	Southernmost latitude
Greater Cape Floristic Region, South Africa	18.0121	23.4480	-32.1368	-34.8096

GCMD keywords: BIOSPHERE > VEGETATION > VEGETATION COVER; BIOSPHERE > VEGETATION > VEGETATION SPECIES; AFRICA > SOUTHERN AFRICA > SOUTH AFRICA; platform FIELD SURVEYS; instrument GPS.

File naming convention

`bioscape_veg_{level}_{table}_v{YYYYMMDD}.csv`, `bioscape_veg_line_intercepts_v{YYYYMMDD}.csv`, and `bioscape_veg_spatial_v{YYYYMMDD}.gpkg`

- `level` is `plot` or `quadrat`.
- `table` is `summary` (one row per location) or `species` (one row per taxon at that location).
- `line_intercepts` is the point-intercept table along the plot diameters.

- YYYYMMDD is the date these version 1 files were written. This submission is v20261007. The vegetation sources are v20261007 and previously released to the team on GitHub. The spatial source is v20261002. Publication at ORNL is version 1. There is no earlier DOI.

Names use lowercase letters, numbers, and underscores.

File descriptions

The CSV files have one header row. Numeric missing values are -9999. Text missing values are NA. **record_id** is unique within each CSV.

file	rows	bytes	sha256
bioscape_veg_plot_summary_v20261007.csv	189	22006	4005ed742f00b909aa7c7c4a89937cc222a6d48b2718d61e3010c2a4a4aaffc2
bioscape_veg_plot_species_v20261007.csv	1974	207902	f3b4f453fbc9dd53f48b1e23d3735e04fb3725801e6cb6747d6f88c7404599ea
bioscape_veg_quadrat_summary_v20261007.csv	756	162269	98c2d1b63deb05e4533f405b6e93a291e5ea84f5137e1ca7a5fa3c00516da1e7
bioscape_veg_quadrat_species_v20261007.csv	4795	781188	67f1cc3395fbd2a6bcfdab245821592a31759f603dad9632c548cf8606ea8ae3
bioscape_veg_line_intercepts_v20261007.csv	8316	938734	0f14f16b252b6d3f0d85a08c88011348c384f0255e8d63001a4ba1bf99631b0c
bioscape_veg_spatial_v20261007.gpkg	499	2191360	5ee6ce7b540e842f3394cdb93e92789844e5532392536ca50ee0c7343462a46a

- **Plot summary** (189 rows). One row per surveyed plot. Fractional cover, vegetation height, and soil depth are means of the four quadrants. **site_code_plot** is unique.
- **Plot species** (1974 rows). One row per accepted taxon per plot. Live and dead cover are the sum of the quadrant percent covers divided by 4. Counts are sums across quadrants. **site_code_plot** plus **genus_species_combo** is unique except where noted in Quality assessment.
- **Quadrat summary** (756 rows). One row per quadrant. **site_code_plot_quadrant** is unique. All 189 plots have NE, NW, SE, and SW.
- **Quadrat species** (4795 rows). One row per taxon recorded in a quadrant, after identical rows were dropped. The plot species file is the aggregate of the source table, not of this file after duplicate removal.
- **Line intercepts** (8316 rows). One row per intercept hit (a plant or other cover type) at the left or right of each 1 m mark on the north-south (NS) and west-east (WE) diameters. Most plots have 44 hits (22 per transect). Plot T086 has no line-intercept rows. 1 plot has two hits at every mark.

Spatial file. bioscape_veg_spatial_v20261007.gpkg is a GeoPackage in WGS 84 (EPSG:4326). Layers are **plot_centers**, **plot_parking**, and **plot_polygons**.

- **plot_centers**, 189 points and 189 plot ids. No plot id is repeated. Center ids absent from the plot summary: none. **bioscape_plot_id** matches **plot**.
- **plot_parking**, 122 points where a vehicle was parked for the survey.
- **plot_polygons**, 188 Patches of vegetation the botanists deemed to be representative of the plot (e.g. similar species, post-fire age, soil) that are at least 15 meters in radius and ideally over 50 meters.

plot_centers attributes:

Variable	Description
bioscape_plot_id	Name of the BioScape plot. Matches plot .
region	General region where the plot is located.
botanist	Name of the botanist who conducted the vegetation survey.
date_time	Clock time of the GPS measurement, YYYY-MM-DDThh:mm:ss. The source text did not include a timezone.
name	Name the botanist gave the plot, kept for provenance.
plot_note	Notes made by the botanist, kept for provenance.

Variable	Description
description	Short free-text note.
quality_flag	Flag assigned by the Townsend team while visiting the plot during BioSCape. Values are listed below.
quality_flag_2	A second Townsend visit flag. One plot is Boundary .
location_flag	The original GPS may not have been taken at the plot center. This flag is set when that plot's BioSCape center and its own Townsend center differ by more than 10 m. The stored value is See Townsend alternative location . See townsend_long and townsend_lat .
pt_plot_a	Townsend opportunistic plot associated with this BioSCape plot. Additional plots were collected where the original plot had a quality flag, where nearby vegetation was similar but had a different dominant species, or where a nearby patch was a single species.
pt_plot_b	A second Townsend plot label. See pt_plot_a .
pt_plot_c	A third Townsend plot label. See pt_plot_a .
pt_plot_d	A fourth Townsend plot label. See pt_plot_a .
townsend_long	Alternative longitude, decimal degrees, of the plot center marked by the Townsend team.
townsend_lat	Alternative latitude, decimal degrees, of the plot center marked by the Townsend team.
geom	Longitude and latitude of the BioSCape plot center.

quality_flag values in plot_centers:

quality_flag	Plots
Low vegetation	24
Boundary	10
Other	7
Trampled	3
Accessibility	2
Removed invasive	2
Grazing	1

Boundary means the plot was placed on the boundary of two vegetation types. The Townsend team then set up additional plots nearby to record the distinct communities (**pt_plot_a** through **pt_plot_d**). **Low vegetation** means cover was very low, and an alternative plot was often marked nearby where cover was higher. **Trampled** means previous visitors had trampled the plot. **Removed invasive** means a Townsend-team botanist removed an invasive individual that could have been present during the flights. **Grazing** means there was evidence of grazing. **Other** and **Accessibility** are stored on some plots.

plot_parking columns are bioscape_plot_id, region, name, description, and date_time.
plot_polygons columns are bioscape_plot_id, region, botanist, name, description, and date_time.

Code. workflow/plot_data_compile_xlsx.R in <https://github.com/BioSCape-io/BioSCape-terrestrial>.

Data file properties

Tabular CSV, plus one GeoPackage. The five tables have no map projection. bioscape_veg_spatial_v20261007.gpkg uses geographic WGS 84 (EPSG:4326). Processing level: in situ field observations, with accepted names applied at compilation. These are not a satellite Level 2–4 product.

Data details

Shared codes:

- quadrant: NE, NW, SE, SW.

- **line_transect**: NS, WE.
- **metres_along_line**: meter mark 0–10 plus L or R for the left or right side of the transect rope (for example 0L, 5R).
- **other_cover_type**: Bare soil, Bare rock, Bare soil and rock, Dead plant, or Other.
- **clonal** and **clonal_yes_no**: yes, no, or NA.
- **seasonally_apparent**: 0, 1, or -9999.
- **groundwater**: Well-drained, Impeded drainage, Seepage, Swamp, Stream bank, Suurvlakke, or NA.
- **sampled** on the plot summary is 1 for every row.
- **inat_id**: iNaturalist observation id. Reconstruct the observation page as https://www.inaturalist.org/observations/{inat_id}.

bioscape_veg_plot_summary_v20261007.csv

Variable	Units	Description
record_id		Unique row number in this file.
plot		Plot code, T plus three digits. Unique in this file.
site_code		Survey area name.
site_code_plot		Site and plot joined by _ . Unique row key.
gps_plot_centre		Text note on how the center was recorded. Not a coordinate.
observer		Botanist named on the field sheet. deb is stored as Douglas Euston-Brown.
percent_bare_soil	percent	Mean of the four quadrant estimates. Missing: -9999.
percent_bare_rock	percent	Mean of the four quadrant estimates. Missing: -9999.
percent_dead_vegetation	percent	Mean of the four quadrant estimates. Missing: -9999.
percent_live_vegetation	percent	Mean of the four quadrant estimates. Missing: -9999.
veg_height_mean_cm	cm	Mean vegetation height across quadrants. Missing: -9999.
soil_depth_cm	cm	Mean soil depth across quadrants. Missing: -9999.
groundwater		Drainage class taken from the quadrants.
access_notes		Free-text access note.
sampled		Completed-survey flag. Always 1.

This file has no survey date. Take **date** from the quadrat summary, joining on **plot**. Field surveys were recorded by Ross Turner, Douglas Euston-Brown, Steven Molteno, Stuart Hall, and Paul Emms.

bioscape_veg_plot_species_v20261007.csv

Variable	Units	Description
record_id		Unique row number in this file.
plot		Plot code.
site_code		Survey area name.
site_code_plot		Site and plot joined by _ .
accepted_genus		Accepted genus.
accepted_species		Accepted species epithet.
name_check		Name-check code copied from the field sheet. The list that issued the numbers is not identified in the source workbooks. Missing: NA.
genus_species_combo		Accepted genus and species as one label.
percent_cover_alive	percent	Sum of quadrant live cover, divided by 4. Missing: -9999.
percent_cover_dead	percent	Sum of quadrant dead cover, divided by 4. Missing: -9999.
abundance_alive_count	count	Sum of live counts across quadrants. Missing: -9999.
abundance_dead_count	count	Sum of dead counts across quadrants. Missing: -9999.
mean_canopy_diameter_cm	cm	Mean of recorded quadrant canopy diameters. Missing: -9999.
clonal		yes, no, or NA.
seasonally_apparent		Maximum of the quadrant values. Missing: -9999.

bioscape_veg_quadrat_summary_v20261007.csv

Variable	Units	Description
record_id		Unique row number in this file.
plot		Plot code.
site_code		Survey area name.
site_code_plot		Site and plot joined by _.
site_code_plot_quadrant		Site, plot, and quadrant joined by _ . Unique row key.
quadrant		NE, NW, SE, or SW.
gps_plot_centre		Text note. Not a coordinate.
observer		Botanist named on the field sheet.
date		Survey date, YYYY-MM-DD.
percent_bare_soil	percent	Quadrant estimate. Missing: -9999.
percent_bare_rock	percent	Quadrant estimate. Missing: -9999.
percent_dead_vegetation	percent	Quadrant estimate. Missing: -9999.
percent_live_vegetation	percent	Quadrant estimate. Missing: -9999.
total_cover_check	percent	Sum of the four cover fractions. 14 quadrants are not 100 and are not missing. Missing: -9999.
veg_height_cm	cm	Vegetation height. Missing: -9999.
post_fire_age_years	text	Time since fire as written on the sheet. The column is text because some entries are notes or Excel date serials rather than ages.
soil_depth_cm	cm	Soil depth. Missing: -9999.
groundwater		Drainage class.
parking_gps		Text note that a parking location was recorded. Not a coordinate.
access_notes		Free-text access note.
comments		Free-text site description.
sheet_name		Source workbook: Ross01, Ross01b, Ross01c, Bio02, Bio03, or Bio04.
old_plot		Plot number before the Swartberg renumbering in Methods. Missing: -9999.

bioscape_veg_quadrat_species_v20261007.csv

Variable	Units	Description
record_id		Unique row number in this file.
plot		Plot code.
site_code		Survey area name.
site_code_plot		Site and plot joined by _.
site_code_plot_quadrant		Site, plot, and quadrant joined by _.
quadrant		NE, NW, SE, or SW.
accepted_genus		Accepted genus.
accepted_species		Accepted species epithet. NA when the record is genus only.
subspecies_or_variant		Subspecies or variant when one was recorded.
genus_species_combo		Genus and species label.
name_check		Name-check code. NA on 164 of 4795 rows.
new_species		Field note for a new or uncertain taxon. iNaturalist URLs are stored as inat_id . Remaining values can be notes or times. yes, no, or NA .
clonal_yes_no		
mean_canopy_diameter_cm	cm	Canopy diameter of the taxon in the quadrant. Missing: -9999.
abundance_alive_count	count	Live individuals. Missing: -9999.
percent_cover_alive	percent	Live cover in the quadrant. Missing: -9999.
percent_cover_dead	percent	Dead cover in the quadrant. Missing: -9999.
abundance_dead_count	count	Dead individuals. Missing: -9999.
seasonally_apparent		0, 1, or -9999.
comment		Note on the taxon in that quadrant. iNaturalist URLs are stored as inat_id ; other residual text is kept. yes, no, or NA .
clonal		
taxon		Same label as genus_species_combo .
old_plot		Plot number before Swartberg renumbering. Missing: -9999.
inat_id		iNaturalist observation id extracted from new_species or comment . Missing: NA.

bioscape_veg_line_intercepts_v20261007.csv

Variable	Units	Description
record_id		Unique row number in this file.
plot		Plot code.
site_code		Survey area name.
site_code_plot		Site and plot joined by _.
site_code_plot_line_transect		Site, plot, and transect joined by _.
line_transect		NS or WE.
metres_along_line		Meter mark and side of the rope, for example 0L or 10R.
accepted_genus		Accepted genus when the hit is a plant.
accepted_species		Accepted species epithet. NA when the record is genus only or a non-plant hit.
subspecies_or_variant		Subspecies or variant when one was recorded.
genus_species_combo		Genus and species label. NA for non-plant hits.
name_check		Name-check code. Missing: NA.
other_cover_type		Non-plant cover at the intercept when no taxon was recorded, or when both were noted.
comments		Free-text note. iNaturalist URLs are stored as inat_id .
old_plot		Plot number before Swartberg renumbering. Missing: -9999.
inat_id		iNaturalist observation id extracted from comments . Missing: NA.

Application and derivation

These plots are the environmentally stratified vegetation surveys for BioSCape. They are the ground observations for comparing field composition and structure with the campaign airborne imaging spectroscopy and lidar (AVIRIS-NG, PRISM, LVIS, and HyTES) over the Greater Cape Floristic Region (Cardoso et al. 2025). Use the plot species table for one cover and count per taxon per plot. Use the quadrat tables when the within-plot measurements matter. Use the line intercepts table for point hits along the plot diameters. The woody-invasive plots (ORNL DAAC <https://doi.org/10.3334/ORNLDAAAC/2511>) were sited near this network and are comparable only after that placement difference is taken into account.

Quality assessment

Cover fractions are visual estimates. This version does not add an uncertainty column. **total_cover_check** is the field sum of bare soil, bare rock, dead vegetation, and live vegetation. Quadrants that do not sum to 100:

site_code_plot_quadrant	total_cover_check
capepoint_T070_NE	104
capepoint_T070_NW	105
capepoint_T070_SE	102
capepoint_T070_SW	105
capepoint_T265_NE	120
capepoint_T265_NW	130
capepoint_T265_SE	150
capepoint_T265_SW	140
gardenroute_T148_SW	101
langeberg_T117_NE	107
langeberg_T117_NW	105
outeniqua_T261_NW	90
rooiberg_T003_SW	103
swartberg_T026_NW	115

Version 1 changes from the source tables:

- Site codes on 0 plots in the species tables were replaced with the plot-summary spelling so **site_code_plot** joins.
- 1 identical quadrat-species rows were removed.

- iNaturalist observation URLs in `new_species`, `comment`, and `line_comments` were moved to `inat_id` (37 quadrat-species rows and 20 line-intercept rows had an iNaturalist URL). Residual free-text notes were kept. Non-iNaturalist URLs were left in place.
- Personal contact details in `access_notes` were redacted.
- The observer value `deb` and the spatial botanist name `Doug Euston-Brown` are stored as `Douglas Euston-Brown`.
- Spatial source v20261002 drops the duplicate T114 center and its location flag, sets T074 region to `CapePoint`, links T183 to associated plot `pt183`, stores `pt_plot_d` as one column, and uses real missing values instead of the text `NA` in Townsend plot labels.

Relative to the data pre-release v20241104, the v20261007 sources rename T222 *Senecio subcanescens* to *Senecio rigidus* and set T091 *Metalasia densa* live abundance from 20 to 1 (with a note on the inconsistent cover versus count).

13 quadrat-by-taxon keys still occur more than once because the cover or count differed. `record_id` keeps those rows distinct. The plot species file was summed from the source table, so those repeated measurements are included in its cover and counts.

site_code_plot_quadrant	genus_species_combo	n
anysberg_T106_SE	Elytropappus rhinocerotis	2
anysberg_T108_NW	Ruschia pungens	2
cederberg_T243_SW	Anthospermum aethiopicum	2
dehoop_T132_NE	Pterocelastrus tricuspidatus	2
gardenroute_T147_NE	Helichrysum patulum	2
gardenroute_T147_NW	Helichrysum patulum	2
gardenroute_T147_SE	Helichrysum patulum	2
gardenroute_T147_SW	Helichrysum patulum	2
langeberg_T186_SE	Erica anguliger	2
langeberg_T186_SE	Penaea mucronata	2
outeniqua_T185_SE	Protea neriifolia	2
rooiberg_T003_SW	Drosanthemum	2
rooiberg_T245_SW	Agathosma mundtii	2

Line intercepts are present for 188 of 189 plots (missing: T086). Plot T096 has two hits at every transect mark (88 rows). `post_fire_age_years` remains text. `name_check` is the code written on the field sheet; the authority list is not in these tables.

Data acquisition, materials, and methods

Plots were placed across reserves and passes in the Greater Cape Floristic Region for the 2023 BioScape field campaign (Cardoso et al. 2025). Each surveyed plot has four quadrants: NE, NW, SE, and SW. In each quadrant the botanist recorded fractional cover of bare soil, bare rock, dead vegetation, and live vegetation; vegetation height; soil depth; drainage; and, for each vascular plant taxon, live and dead percent cover, live and dead counts, and canopy diameter. Centers were located with a Trimble GPS. The coordinate values are in the `plot_centers` layer of bioscape_veg_spatial_v20261007.gpkg.

Two perpendicular line transects cross at the plot center along the north–south (NS) and west–east (WE) diameters. At each 1 m mark from 0 to 10 m, the botanist recorded the plant or other cover immediately left (L) and right (R) of the rope from a canopy view. Those hits are stored in the line-intercepts table.

The version 1 files were written by this document. Column names are snake case. Numeric gaps are `-9999` and text gaps are `NA`. Survey dates are `YYYY-MM-DD`. GPS times are `YYYY-MM-DDThh:mm:ss` with the clock time taken from the source string. Each CSV has `record_id`. Clonal codes are `yes` or `no`. `old_plot` is the plot number before renumbering.

Field records were kept in six Google Sheets workbooks (Ross01, Ross01b, Ross01c, Bio02, Bio03, Bio04). The v20261007 compile was built from workbook downloads dated 2026-10-07.

`workflow/plot_data_compile_xlsx.R` stacked the SiteData tabs, plot tabs, and line-intercept tabs, and rebuilt the plot and quadrant identifiers. Swartberg plot numbers that collided with numbers used at other sites were renumbered. The previous number is `old_plot`. The source script maps Swartberg 20 to T110, 22 to T012, 23 to T013, and 24 to T014.

Plot summary cover, height, and soil depth are the mean of the four quadrants. Plot species live and dead cover are the sum of that taxon’s quadrant covers divided by 4, and counts are sums. Unrecorded quadrant occurrences add nothing to the sum. With four quadrants in every released plot, dividing by 4 is that taxon’s mean contribution to the plot.

References

- Cardoso, A.W., E.L. Hestir, J.A. Slingsby, C.J. Forbes, G.R. Moncrieff, W. Turner, A.L. Skowno, J. Nesslage, P.G. Brodrick, K.D. Gaddis, and A.M. Wilson. 2025. The biodiversity survey of the Cape (BioSCape), integrating remote sensing with biodiversity science. *npj Biodiversity* 4:2. <https://doi.org/10.1038/s44185-024-00071-5>
- Fitzpatrick, M.C., D. Spalink, A.J. Elmore, J. Measey, and S. Kritzing-Klopper. 2026. BioSCape: Vegetation Composition at Plots with Woody Invasives, CFR, South Africa. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORN LDAAC/2511>
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Dataset revisions

This release is compiled from v20261007 vegetation tables and the v20261002 spatial GeoPackage from the team GitHub repository. It was updated to conform to ORNL data standards as v20261007. This is the first submission of these tables to ORNL DAAC. There is no prior DOI.